

Means Separation and Data Presentation

AGRON 5130

YOUR NAME HERE

Introduction

The goal of this assignment is to give you hands-on practice with **means separation**, **linear contrasts**, and **presenting treatment means**.

You will work with an apple yield dataset to practice:

- fitting an ANOVA model
- comparing treatment means using **LSD** and **Tukey** comparisons
- constructing and testing **linear contrasts**
- presenting treatment means in a **figure with grouping letters**

Submit a **knitted PDF/HTML** generated from your **R Markdown** file to Canvas.

We can start by loading the necessary libraries

```
library(ggplot2)
library(emmeans)
library(multcomp)
library(nlme)
```

Question 1

1 point

Load the dataset "data/apple.csv" and display the first few rows of the dataset. This dataset contains apple yield data from an experiment that evaluated foliar treatments. The experiment was conducted as a randomized complete block design with 4 replications.

```
apple <- read.csv('./data/apple.csv')
head(apple)
```

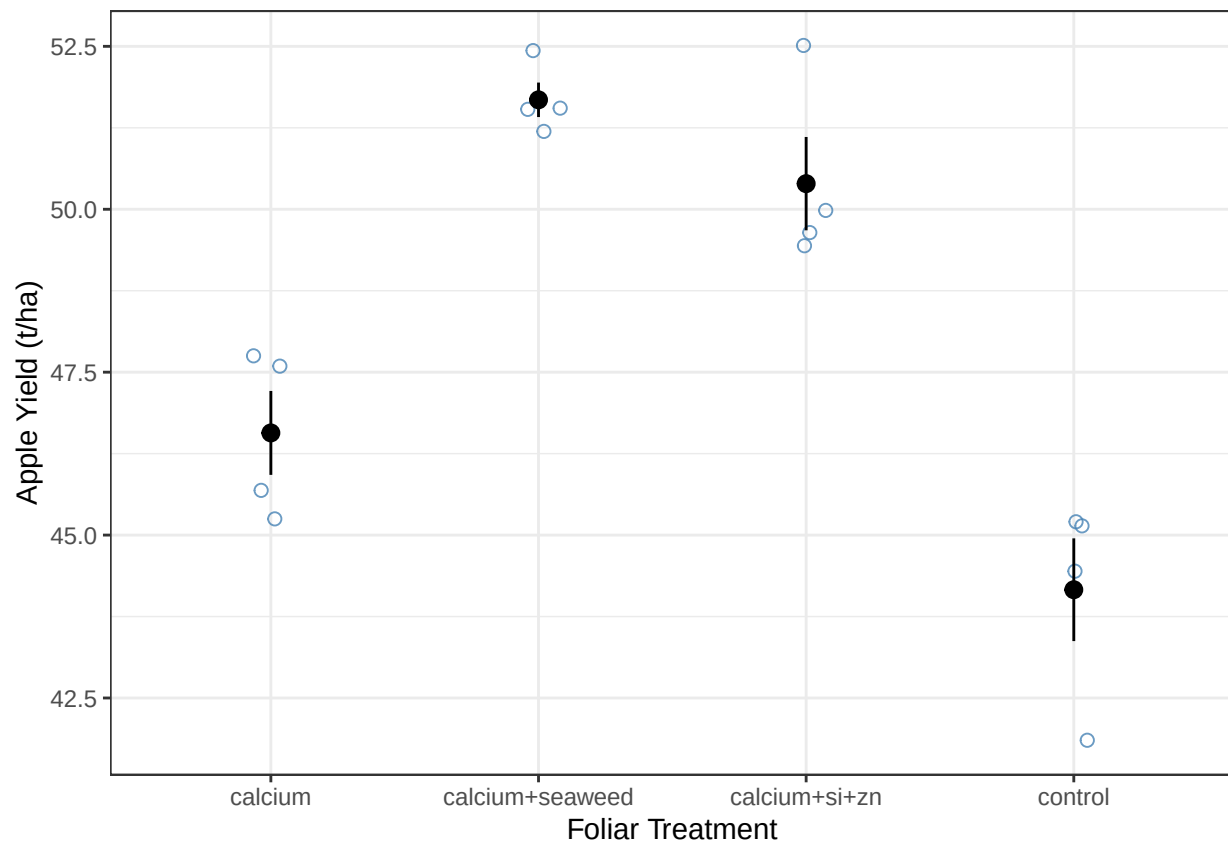
```
##   block    foliar_trt   yield
## 1     1 calcium+si+zn 49.43409
## 2     1 calcium+seaweed 51.55156
## 3     1          calcium 47.75001
## 4     1          control 45.13852
## 5     2 calcium+si+zn 52.51426
## 6     2 calcium+seaweed 52.43407
```

Question 2

2 points

Build a plot to visualize the relationship between yield and foliar_trt

```
ggplot(apple, aes(x = foliar_trt, y = yield)) +  
  geom_jitter(width = 0.1,  
             size = 2,  
             alpha = 0.8,  
             color = "steelblue",  
             shape = 1) +  
  labs(x = "Foliar Treatment",  
       y = "Apple Yield (t/ha)") +  
  stat_summary(fun.data = 'mean_se') +  
  theme_bw()
```



Question 3

2 points

Fit an ANOVA model testing the effect of foliar_trt on apple yield.

Show the ANOVA table.

```
model <- lme(yield ~ foliar_trt,
             random = ~ 1 | block,
             data = apple)
anova(model)
```

```
##              numDF denDF  F-value p-value
## (Intercept)      1      9 16845.27 <.0001
## foliar_trt       3      9   33.63 <.0001
```

Question 4

2 points

Use the **emmeans** package to compute the estimated marginal means for **foliar_trt**.
Then conduct **pairwise comparisons using LSD inference**.

Show the output.

```
emm <- emmeans(model, ~ foliar_trt)
pairs(emm, adjust = 'none')
```

```
## contrast              estimate      SE df t.ratio p.value
## calcium - (calcium+seaweed)    -5.11 0.843  9  -6.060 0.0002
## calcium - (calcium+si+zn)     -3.83 0.843  9  -4.535 0.0014
## calcium - control              2.41 0.843  9   2.853 0.0190
## (calcium+seaweed) - (calcium+si+zn)  1.29 0.843  9   1.525 0.1616
## (calcium+seaweed) - control      7.52 0.843  9   8.913 <.0001
## (calcium+si+zn) - control       6.23 0.843  9   7.388 <.0001
##
## Degrees-of-freedom method: containment
```

Question 5

3 points

Now conduct the same pairwise comparisons using **Tukey's HSD adjustment**.

Show the results and briefly explain how they differ from the LSD-style comparisons.

```
pairs(emm, adjust = 'tukey')
```

```
## contrast              estimate      SE df t.ratio p.value
## calcium - (calcium+seaweed)    -5.11 0.843  9  -6.060 0.0009
## calcium - (calcium+si+zn)     -3.83 0.843  9  -4.535 0.0063
## calcium - control              2.41 0.843  9   2.853 0.0749
## (calcium+seaweed) - (calcium+si+zn)  1.29 0.843  9   1.525 0.4628
## (calcium+seaweed) - control      7.52 0.843  9   8.913 <.0001
## (calcium+si+zn) - control       6.23 0.843  9   7.388 0.0002
##
## Degrees-of-freedom method: containment
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Question 6

3 points

Now test a **linear contrast** comparing
calcium plus additional ingredients

versus

calcium alone.

First examine the order of treatment levels in `foliar_trt`. Then compute the estimated marginal means using `emmeans`.

Construct the appropriate **contrast coefficients** and test the hypothesis using the `contrast()` function.

Briefly interpret the results.

```
contrast(emm,
  method = list(
    calcium_vs_alone = c(-1, -1/2, -1/2, 0)
  ))

## contrast      estimate      SE df  t.ratio p.value
## calcium_vs_alone    -97.6 0.854  3 -114.274  <.0001
##
## Degrees-of-freedom method: containment
```

Question 7

1 point

Use `cld()` to obtain **Tukey grouping letters** for the treatment means.

Show the output.

```
cld(emm, adjust = 'tukey', Letters = letters)

## Note: adjust = "tukey" was changed to "sidak"
## because "tukey" is only appropriate for one set of pairwise comparisons

## foliar_trt      emmean      SE df lower.CL upper.CL .group
## control         44.2 0.636  3    40.8    47.6    a
## calcium          46.6 0.636  3    43.2    50.0    a
## calcium+si+zn    50.4 0.636  3    47.0    53.8    b
## calcium+seaweed  51.7 0.636  3    48.3    55.1    b
##
## Degrees-of-freedom method: containment
## Confidence level used: 0.95
## Conf-level adjustment: sidak method for 4 estimates
## P value adjustment: tukey method for comparing a family of 4 estimates
## significance level used: alpha = 0.05
## NOTE: If two or more means share the same grouping symbol,
##       then we cannot show them to be different.
##       But we also did not show them to be the same.
```

Question 8

1 point

Create a **plot using ggplot2** that shows:

- treatment means
- confidence intervals
- grouping letters above the bars

Your final figure should resemble the type of plot shown in the lecture.